

# **Hotel Booking Cancellation Prediction**

**A Business Intelligence Solution for Revenue Management**

## **Final Project Report**

BIA 5450 Capstone Project

Humber Polytechnic

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## 1. Executive Summary

This project delivered a business intelligence solution that predicted which hotel bookings were likely to be cancelled and presented the result to a revenue manager as money at risk and a ranked list of bookings to act on. The team worked through four stages: business problem identification and scope analysis, data preparation, solution design, and solution implementation with optimization, combining a Python machine learning pipeline with a Microsoft Power BI reporting layer. The source was the public Hotel Booking Demand dataset, which held 119,390 bookings from two Portuguese hotels with arrivals between July 2015 and August 2017. Data preparation retained 118,563 valid records under documented decision rules and separated 827 invalid records into an audit file rather than deleting them. Fields that revealed the booking outcome, and fields that only exist after a booking is made, were removed so the model could be used on a new booking at the moment it arrives.

Three algorithms were compared on a temporal split, training on 94,850 earlier bookings and testing on 23,713 later ones. The final model was gradient boosting (optimized), which reached a ROC-AUC of 0.8798 on the held out period against a target of 0.80. Operated at a decision threshold of 0.25 rather than the default, it caught 82.3 percent of the bookings that really cancelled, and 68.6 percent of the bookings it flagged did cancel. The largest single improvement did not come from tuning. Removing one calendar feature that could not generalise beyond the months it was trained on moved gradient boosting from 0.8238 to 0.8798, from the weakest of the three models to the strongest. Across the 23,713 bookings in the test period, which carried a total booking value of \$12,101,899, the solution measured \$4,275,959 of Revenue at Risk and produced an action list of 11,656 bookings carrying \$3,646,308 of that exposure, each with a risk band and a recommended action attached.

Testing was not limited to model metrics. A control total test compared the nine dashboard measures against the scoring output and found that the first version of the dashboard had been built on two tables covering two different populations, which produced a headline cancellation rate of roughly 8 percent against an actual test period rate of 40.97%. The reporting layer was rebuilt on a single table and every measure now reconciles. The team recommends that the hotel adopt the ranked action list as the daily working queue for the revenue desk, and that any

production deployment retrain the model on current data from its own property management system before the recommended actions are trusted operationally.

## 2. Introduction

This report will describe the complete capstone project, from the identification of the business problem through to the optimization of the delivered solution. It is organised to follow the four sprints of the capstone process, and each section will build on the one before it, so that a reader who starts at the beginning will be able to follow how a raw booking file became a decision a revenue manager can act on.

Section 3 will set out the business problem and explain why hotels lose money to cancellations and why the current industry practice of a fixed overbooking percentage does not solve it. Section 4 will present the analytics questions, covering descriptive, diagnostic, predictive and prescriptive types, and will tie each question to the decision it enables. Section 5 will state the scope of the project, including what the team deliberately excluded and the constraints the team worked under.

Section 6 will describe the data source, the business entities it contains, the systems that would hold those entities in a live hotel, and the flow of data through the solution. Section 7 will explain the data manipulation process, the decision rules used to handle missing values, duplicates and invalid records, and the three prepared outputs the pipeline produces.

Section 8 will present the solution design and explain how the new analytical layer fits between the booking data source and the reporting layer without disturbing the systems a hotel already depends on. Section 9 will report the implementation and the outcome testing, including the comparison of three algorithms, the results against the target outcomes, the business test scenarios, and the control total test applied to the dashboard. Section 10 will describe the optimization work, which moved the solution from its first working version to its final one, and will identify what remains to be optimized. Section 11 will conclude, and the appendices will hold the glossary, the individual contribution log, the model drivers, the dashboard measures and the register of files the project produced.

### 3. Business Problem Overview

Hotels need to know which bookings are likely to cancel while there is still time to act, and in most properties today they cannot. Between 15 and 40 percent of hotel bookings are cancelled before the guest arrives, and the proportion varies sharply by booking channel and market segment (Antonio, de Almeida, & Nunes, 2017). When a cancellation lands close to the arrival date, the room often cannot be re-sold at any price, so a late cancellation is not deferred revenue, it is lost revenue.

The standard industry response is a static overbooking percentage, for example selling 105 percent of inventory during a peak season. That approach is blunt in both directions. It treats every reservation as carrying the same risk, so it leaves money on the table when the book is genuinely firm, and it produces walked guests when the book is weaker than the average. A booking made two hundred days ahead through an online travel agent with no deposit does not behave like a booking made five days ahead directly with the hotel with a deposit attached, yet a single percentage cannot tell them apart.

The data management position in a typical hotel makes this harder. Booking records are spread across a property management system, a central reservation system and several online travel agent extranets, and where they are consolidated they are used descriptively, to report what already happened, rather than predictively. There is no per booking risk score, no translation of that risk into money, and no ranked view of which reservations deserve attention this week, so decisions about overbooking and deposit policy are made on rules of thumb rather than on data the hotel already owns.

The gap this project addressed was therefore twofold. First, a per booking probability of cancellation that uses only information available at the moment of booking. Second, a way of putting that probability in front of a revenue manager in the terms they actually work in, which are dollars of exposure and a ranked list of reservations, not a probability between zero and one. The stakeholders for that output are listed in Table 1, with the revenue manager as the primary daily user and four secondary roles that consume specific views.

Table 1. Stakeholders and how each one uses the solution.

| Stakeholder        | Role and concern                                                           | How they use the solution                                                      | Interest          |
|--------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------|
| Revenue Manager    | Primary user, maximises revenue per available room and manages overbooking | Works the ranked action list daily and sets overbooking levels two weeks out   | High, daily       |
| General Manager    | Owens the top line revenue and occupancy targets                           | Watches the Revenue at Risk indicator and uses it for ownership reporting      | High, weekly      |
| Marketing Manager  | Owens channel performance and customer retention                           | Compares channel profitability after cancellations and targets retention spend | Medium, monthly   |
| Front Desk Manager | Owens operational smoothness and walk in handling                          | Uses the high risk arrivals view to manage held rooms                          | Medium, daily     |
| Director of Sales  | Owens group bookings, where a single cancellation is large                 | Reviews group bookings ranked by risk and opens deposit conversations early    | Medium, per group |

Cancellation is a problem across the hospitality industry rather than one specific to the two hotels in the dataset, so the method transfers to other properties. What would change is the data feed and the retrained model, not the design of the solution.

## 4. Analytics Questions

The analytics questions run from descriptive to prescriptive, in the order a revenue team actually works through the problem. Table 2 lists them by type and states the decision each one enables, because a question that does not change a decision does not earn its place in the dashboard.

Table 2. Analytics questions by type and the decision each one enables.

| Type         | Analytics question                                                                                  | Decision it enables                                                     |
|--------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Descriptive  | What is our cancellation rate by quarter, hotel and market segment?                                 | Establishes the baseline level of loss and shows where it concentrates  |
| Descriptive  | Which channels and countries contribute the most cancellations in absolute terms?                   | Focuses attention on the highest volume sources of loss                 |
| Diagnostic   | Why do online travel agent bookings cancel far more often than direct bookings?                     | Points to channel commitment as a deposit policy lever                  |
| Diagnostic   | Why does cancellation behaviour differ so sharply across deposit types?                             | Tests whether deposit policy itself changes guest behaviour             |
| Predictive   | Given a new booking, what is the probability that it will be cancelled?                             | Produces the per booking risk score that drives every downstream action |
| Predictive   | Across the bookings on the books, how many are likely to cancel and what is the exposure?           | Sizes the forward exposure for overbooking and staffing decisions       |
| Prescriptive | Which bookings should receive a retention offer this week, and which should be asked for a deposit? | Generates the daily action list, with the action set by risk band       |
| Prescriptive | What overbooking level should be set for a given arrival date?                                      | Converts the forecast into an inventory decision                        |

The two diagnostic questions were answered during exploration and their answers shaped the design. Deposit type and country turned out to be the two strongest drivers of cancellation risk in the final model, ahead of lead time, which is reported in Section 9. The predictive questions are answered by the model itself, and the prescriptive questions are answered by the business rules layer, which converts a probability into a risk band and a named action.

## 5. Scope Statement

The scope was set to fit a fourteen week term and a four person team, and to deliver something a revenue manager could actually use rather than a partial version of something larger. In scope is everything needed to move a booking from a raw record to a recommended action. Out of scope is anything that adds technical complexity without adding decision value at this stage.

*Table 3. Project scope boundaries.*

| In scope                                                           | Out of scope                                                   |
|--------------------------------------------------------------------|----------------------------------------------------------------|
| Binary cancellation prediction on the Hotel Booking Demand dataset | Multi class no show or late arrival prediction                 |
| Two Portuguese hotels, one city and one resort, 2015 to 2017       | Hotels outside Portugal, or data after 2017                    |
| Logistic regression baseline plus two tree based models            | Deep learning and neural network approaches                    |
| Power BI dashboard for the revenue manager                         | Mobile application or real time push notification              |
| Manually triggered pipeline run and dashboard refresh              | Live retraining on streaming booking data                      |
| Recommended actions by risk band                                   | Automated execution of those actions in the reservation system |
| Revenue at Risk measured in dollars for the test period            | Measured business outcome of the retention offers themselves   |

Three constraints shaped the work. The dataset is historical and covers two properties in one country, so the cancellation percentages reported here are properties of this dataset rather than current market rates. No live database or paid infrastructure was available, so the analytical layer hands off to the reporting layer through a controlled file rather than a database connection. And the team was building its first end to end machine learning solution, which is why the design favours approaches that can be explained and audited over approaches that are marginally more accurate.

Two assumptions underpin the business value figures in Section 10: that contacting a flagged guest costs about 10.0 percent of the booking value, and that a retention action saves about 30.0 percent of the bookings it reaches. Both are planning assumptions taken from published revenue management practice rather than measured results from this hotel, and the report says so wherever a dollar figure depends on them.

## 6. Data Sources, Key Data Entities and Flows

### 6.1 The data source and its credibility

The primary source was the Hotel Booking Demand dataset, supplied as the flat file `hotel_bookings.csv`. It was published as a peer reviewed data article by Antonio, de Almeida and Nunes (2019) and later made available through Kaggle (Mostipak, n.d.), and it describes real booking demand from one resort hotel and one city hotel in Portugal. Table 4 profiles it.

*Table 4. Source data profile.*

| Profile item            | Value                                                 |
|-------------------------|-------------------------------------------------------|
| Source file and format  | hotel_bookings.csv, comma separated values, 16.07 MB  |
| Original size           | 119,390 records, 32 fields                            |
| Hotels represented      | One city hotel and one resort hotel, both in Portugal |
| Arrival period          | July 2015 to August 2017                              |
| Target field            | is_canceled                                           |
| Licence                 | Open, published for research use                      |
| Primary processing tool | Python with pandas                                    |

The source was judged credible on three grounds: the peer reviewed data article speaks to its authority, it carries a clearly defined cancellation target alongside booking, guest, stay, distribution and price attributes, and it is long enough to support a temporal split. Its weaknesses are timeliness and geographic concentration, so a production deployment would need recent data from the client hotel and validation against that hotel's own definitions.

### 6.2 Key data entities and systems of record

The booking is the central business entity, and every row in the source describes one booking, carrying attributes that belong to five related entities: the booking itself, the hotel and stay, the guest, the distribution path, and the outcome. Table 5 names each one and the system that would hold it in a live hotel, which is what makes the design portable to a real property.

*Table 5. Key business entities and their systems of record.*

| Entity         | Representative attributes                                                | System of record in production                                |
|----------------|--------------------------------------------------------------------------|---------------------------------------------------------------|
| Booking        | Lead time, arrival date, deposit type, customer type, booking value      | Property management system                                    |
| Hotel and stay | Hotel, nights, meal plan, room types, average daily rate                 | Property management system                                    |
| Guest          | Guest counts, country, repeat history, special requests                  | Property management system and customer relationship platform |
| Distribution   | Market segment, distribution channel, agent, company                     | Central reservation system and online travel agent extranets  |
| Outcome        | Cancellation flag, reservation status, status date                       | Property management system                                    |
| Model output   | Cancellation probability, risk band, Revenue at Risk, recommended action | Created by the analytical layer built in this project         |

The last row is the one the project adds. Everything above it already exists in a hotel that runs a property management system, so the design question was where the scored booking gets created and how it reaches the people who act on it.

### 6.3 Data flow

Figure 1 shows the process and the data together. Each row names a process step, the data it consumes, the data it produces, and the role that owns it. Reading the two middle columns downward gives the data flow on its own, which is why the two are drawn as one diagram rather than two.



Steps 1 to 10 run in sequence: each step waits for the data the previous step produces. Step 11 is conditional, the action depends on the risk band.

*Figure 1. Combined process and data flow, from the raw booking file to the revenue manager action.*

The flow begins with the untouched source file, which is read only for the whole project. Python profiles it for missing values, duplicate looking records, date validity, business rule violations, outliers and target leakage, then standardization, rejection rules and feature engineering produce an analytics dataset and a model ready dataset. That dataset is split on time: the earlier bookings train the model, and the later ones are scored as if they had just arrived. The scoring step produces a cancellation probability, and the business rules step turns it into three things a business user can act on, a Revenue at Risk figure in dollars, a risk band, and a recommended action, all written into one schema controlled table that Power BI reads.

The flow closes on a decision made by a person rather than on a model score, which is what makes this a business intelligence solution rather than a modelling exercise. Step 1 is also the only step tied to the academic data source: in a live hotel it would be a scheduled export from the property management system, and every step after it would be unchanged.

## **7. Data Manipulation Process and Data Output**

### **7.1 What the quality profile found**

The audit ran before any value was modified, so that every later decision could be traced back to evidence. Four fields carried missing values: children with 4 missing, country with 488, agent with 16,340, and company with 112,593. The company field was therefore missing for about 94.31 percent of records and agent for 13.69 percent. Missingness in those two fields was read as potentially meaningful rather than as an error, because an absent agent or company plausibly means that no intermediary or corporate account was involved.

The audit also identified 31,994 excess duplicate looking rows across 40,165 rows belonging to repeated value groups. The source has no natural booking identifier, so an identical row cannot be proven to be an accidental duplicate rather than two rooms booked under the same conditions, and removing them automatically would have destroyed legitimate reservations. They were retained and flagged, which is the first of several places where the team chose traceability over a cleaner looking number. The third finding was target leakage: reservation status and reservation status date state the final outcome of the booking, so a model trained with them would report near perfect accuracy and would be useless on a new booking whose final status is not yet known.

### **7.2 Decision rules**

The rules in Table 6 were written down before any modification was applied. Their purpose was to preserve usable information, to avoid imputation that the data could not support, and to keep an auditable record of anything excluded.

Table 6. Data cleaning decision rules and the reasoning behind each one.

| Issue                           | Decision                                 | Reason                                                                                              |
|---------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| children missing                | Fill with 0 and store as a whole number  | Only four values were missing and zero is the dominant, plausible value                             |
| country missing                 | Replace with Unknown                     | Keeps an otherwise usable booking without inventing a country                                       |
| agent missing                   | Set the code to 0 and create has_agent   | Treats absence as no recorded agent rather than as a numeric value                                  |
| company missing                 | Set the code to 0 and create has_company | Keeps the presence information while excluding an identifier that is 94 percent empty               |
| duplicate looking rows          | Retain and flag                          | No booking identifier exists to prove that a repeated row is an error                               |
| zero guests, or zero nights     | Reject into the audit file               | A valid booking needs at least one guest, and the project analyses overnight room revenue           |
| rate below 0, or above 1,000    | Reject into the audit file               | A negative rate is invalid, and the single value of 5,400 was extreme against a next highest of 510 |
| zero rate with positive nights  | Retain and flag                          | May represent complimentary or special rate activity                                                |
| outcome and post booking fields | Exclude from the model ready output      | Prevents target leakage and the use of information not available at booking time                    |

Applying these rules produced 180 zero guest records, 715 zero night records, one negative rate record, one record above 1,000 and 1,959 zero rate records. Because several conditions overlapped on the same rows, the result was 827 unique rejected rows rather than the sum of the individual counts. Rejected rows were written to a separate file with a rejection reason attached, so a reviewer can see exactly what was excluded and why. Table 7 shows the effect on the dataset as a whole.

Table 7. The dataset before and after preparation.

| Measure            | Before         | After    | Interpretation                       |
|--------------------|----------------|----------|--------------------------------------|
| Records            | 119,390        | 118,563  | 827 separated into the audit file    |
| Columns            | 32             | 48       | 16 engineered and audit fields added |
| Cancelled bookings | 44,224         | 44,175   | 49 cancelled records excluded        |
| Bookings honoured  | 75,166         | 74,388   | 778 records excluded                 |
| Cancellation rate  | 37.04%         | 37.26%   | Target balance stayed stable         |
| Missing cells      | 129,425        | 0        | Handled under the documented rules   |
| Rate range         | -6.38 to 5,400 | 0 to 510 | Invalid and extreme values removed   |

### 7.3 Feature engineering and the model ready dataset

Sixteen fields were derived from the source. The arrival date was assembled from the separate year, month and day fields, and the reservation status date was converted from text to a date type. The engineered analytical fields were total\_guests, total\_nights, is\_family, has\_agent, has\_company, room\_type\_changed, zero\_adr\_flag, booking\_value, arrival\_year\_month, arrival\_quarter, arrival\_season, lead\_time\_group and adr\_group. Of these, booking\_value carries the most weight in the finished solution, because it is the average daily rate multiplied by the number of nights and is therefore the quantity that turns a probability into money.

The model ready dataset was then created by removing two classes of field. The first is outcome leakage: reservation\_status and reservation\_status\_date. The second is post booking information: assigned\_room\_type, booking\_changes, days\_in\_waiting\_list and the raw company identifier. These describe activity that happens after a reservation is created and would not be available when a new booking needs to be scored. The result was 118,563 rows and 37 columns.

### 7.4 Data output and reproducibility

The pipeline writes the three outputs in Table 8. Each one exists for a different consumer, and separating them prevents the reporting layer and the model from quietly disagreeing about which population they describe.

*Table 8. Prepared outputs and their consumers.*

| Output                    | Rows and fields                             | Consumer                                             |
|---------------------------|---------------------------------------------|------------------------------------------------------|
| Cleaned analytics dataset | 118,563 rows, 48 fields                     | Exploration and descriptive analysis                 |
| Model ready dataset       | 118,563 rows, 37 fields                     | Model training and scoring, booking time fields only |
| Power BI base table       | 118,563 rows, 35 fields, schema version 1.1 | The reporting layer, under a fixed schema contract   |
| Rejected records          | 827 rows with a rejection reason            | Audit evidence, never silently deleted               |
| Scored bookings           | 23,713 test period rows                     | The dashboard table, one row per scored booking      |

Prepared files are never edited by hand. Any correction is made in Python and the outputs are regenerated from the untouched source, which is what makes the whole chain reproducible. An end to end reproducibility test reloaded the original file, ran the complete preparation function

and regenerated every output. The independently generated files matched the reference outputs on row count, column count, column order and field values, and reproduced 44,175 cancelled bookings, 74,388 bookings honoured, zero missing values, a minimum of one guest and one night, and a rate range of 0 to 510.

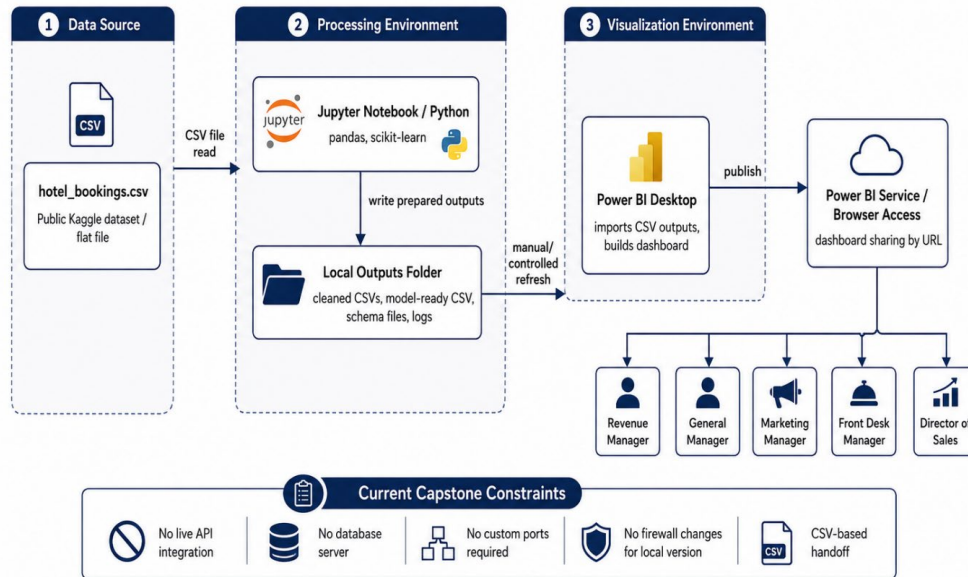
The main challenges in this stage were not the missing values, which were straightforward once the rules were agreed. They were the absence of a booking identifier, which made duplicate handling a judgement call rather than a calculation, the very high missingness in company, and target leakage, which would have produced an impressive and worthless model. In each case the team chose the conservative option and documented it.

## **8. Solution Design and Its Fit into the Existing IT Architecture**

### **8.1 The existing IT architecture**

Figure 2 shows the environment the solution was built in. It is a lightweight analytics environment: a flat file data source, a local Python processing environment, a local output folder, and Power BI for business reporting. It deliberately does not depend on a live database, on application programming interface integrations, or on an automated cloud pipeline, because none of those were available and none of them are needed to demonstrate the full workflow from raw data to business decision.

**Figure 2. Existing IT Architecture (Academic Capstone Version)**



*Figure 2. The existing IT architecture in the capstone environment.*

Table 9 lists the components and the method that connects each pair. The components are joined by a controlled file based workflow: the booking data is processed in Python, validated, and exported under a fixed schema that Power BI reads. File based integration removes the need for database drivers, network ports and interface authentication, while still guaranteeing that the reporting layer only ever reads validated data.

*Table 9. Architecture components and how they connect.*

| Component                           | Role                                                                    | Connection method                                           |
|-------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------|
| hotel_bookings.csv                  | Source of record for the capstone                                       | Read only file read by Python                               |
| Python with pandas and scikit-learn | Validation, cleaning, feature engineering, training and scoring         | Runs locally, writes to the output folder                   |
| Local output folder                 | Holds the prepared datasets, the scored bookings and the control totals | File system, versioned and regenerated only by the pipeline |
| Power BI Desktop                    | Builds the semantic model, the measures and the report pages            | Imports the scored booking table from the output folder     |
| Power BI Service                    | Browser access for stakeholders who do not have Power BI Desktop        | Publish step, available but not yet performed               |

## 8.2 The solution design

The solution uses a hybrid design in which Python carries the data engineering and machine learning layer and Power BI carries the reporting layer. This separation matters for a beginner

team, because it lets the technical work be developed and tested independently of the presentation work, and it means the reporting layer can be rebuilt without retraining anything.

The machine learning design is binary classification. The target is the cancellation flag, where one means the booking was cancelled and zero means it was honoured. The model uses only fields that are available at the time of booking, which is the constraint that drives most of the design decisions described in Section 7. The requirement is to predict whether a booking will cancel while there is still time to act, not to explain cancellations after they have happened.

Three algorithms were compared: logistic regression as an interpretable baseline whose coefficients can be read as risk drivers, and random forest and gradient boosting as alternatives that can capture interactions between booking variables. The evaluation used a temporal split rather than a random split, so the model is always tested on bookings that arrive after the bookings it learned from. A random split would have allowed the model to learn from a booking in August and be tested on a booking in June, which is not how it would be used. Table 10 records why each of these choices was preferred over the alternative the team considered.

*Table 10. Methodology choices and the alternative that was rejected.*

| Decision             | Chosen approach                                                     | Alternative rejected, and why                                                                                |
|----------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Problem framing      | Binary classification of the cancellation flag                      | Regression on days to cancellation, which the data does not support and no stakeholder asked for             |
| Validation design    | Temporal split on arrival date                                      | Random split, which produces an optimistic score by testing on bookings from the same period it trained on   |
| Algorithm family     | One linear model and two tree ensembles, compared on the same split | A single model chosen up front, which would have hidden the fact that the ranking changed after optimization |
| Primary metric       | ROC-AUC, because the solution ranks bookings by risk                | Accuracy, which is misleading when roughly four bookings in ten cancel                                       |
| Business translation | Revenue at Risk in dollars, and a named action per band             | Presenting the probability directly, which no revenue manager acts on                                        |
| Reporting layer      | Power BI reading one schema controlled table                        | Charts generated in Python, which cannot be filtered by a business user                                      |

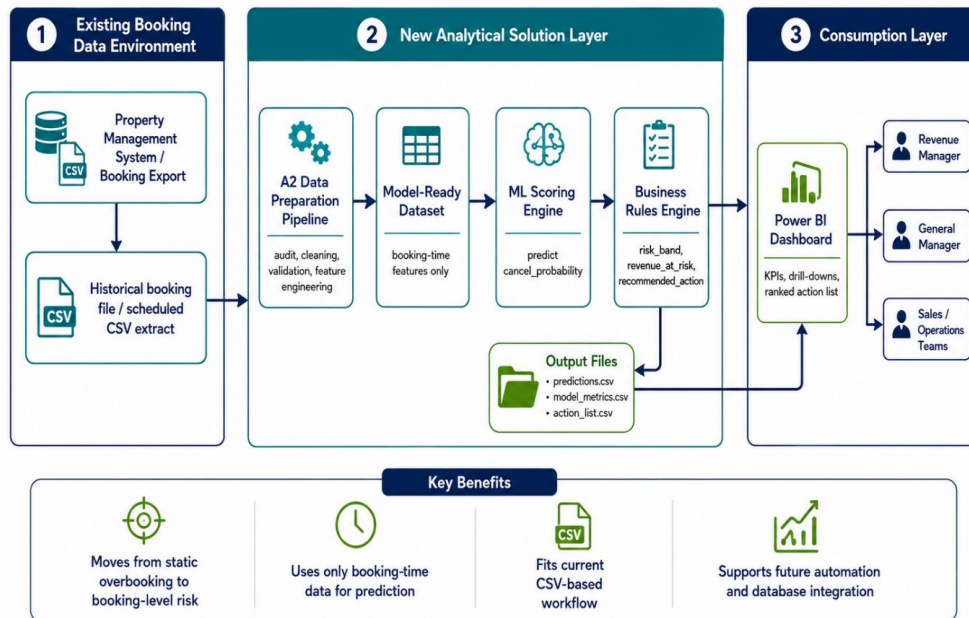
The output of the solution is more than a prediction. The cancellation probability is multiplied by the booking value to give Revenue at Risk, and the probability is mapped to a risk band that carries a recommended action. Expressing the result in currency is what makes the output usable by a revenue manager or a general manager, who work in exposure rather than in

probability. This follows the revenue management literature, which treats cancellation risk as a demand and inventory problem rather than a purely statistical one (Antonio, de Almeida, & Nunes, 2017; Ivanov & Zhechev, 2012).

### 8.3 How the new solution fits the existing architecture

Figure 3 shows the fit. The solution is a new analytical layer that sits between the booking data source and the reporting layer. It does not replace the reservation system or the property management system, and it does not write back to them. It consumes exported booking data and returns cancellation risk intelligence, which means it can be added to a hotel without disturbing the systems that hotel depends on to operate day to day.

**Figure 3. Proposed Solution Fit within the Existing IT Architecture**



*Figure 3. How the new analytical layer fits into the existing IT architecture.*

In the capstone version the existing data environment is represented by the booking file. In production the same role would be played by a scheduled property management system export. The new layer receives that booking data, prepares it through the pipeline described in Section 7, builds the model ready dataset, applies the scoring model, and generates the business rule outputs, which are then written to the table Power BI imports on refresh.

The most important system impact of adding this layer is schema control. Power BI visuals depend on consistent column names and data types, and a renamed column silently breaks a

measure rather than raising an error. Schema version 1.1 fixes the field names, order and types, and reserves the five fields the scoring layer adds: cancellation probability, predicted cancellation flag, risk band, Revenue at Risk and recommended action. Any change to those fields has to be documented and passed to the Visualization Lead before the dashboard is refreshed.

The benefit of this fit is that the hotel moves from a static overbooking rule to booking level risk management without replacing anything it already runs. The revenue manager reviews a ranked list instead of applying one policy to every reservation. The general manager, the marketing manager, the front desk manager and the sales team all read the same numbers, which removes the disagreement that arises when each department maintains its own extract.

The principal risk introduced by the design is over trust. A probability presented on a dashboard can be read as a certainty, and a recommended action can be read as an instruction. The dashboard therefore shows the risk bands, the model performance and the business rules on the same pages as the predictions, so the user can see how confident the model actually is. The solution is designed to support a decision, not to make one.

## **9. Solution Implementation and Outcome Testing**

### **9.1 What was built**

The implementation splits the model ready dataset on arrival date at 2017-04-23, trains on the 94,850 earlier bookings and evaluates on the 23,713 later ones. The cancellation rate differs between the two periods, 36.33% against 40.97%, which is a realistic difficulty rather than a defect: a model deployed in a hotel has to work when behaviour shifts. Categorical and numeric fields pass through separate preprocessing paths inside one pipeline object, so the transformation applied at scoring time is exactly the one fitted at training time, and scoring, business rules and the dashboard handoff all run from that saved object.

### **9.2 Results against the target outcomes**

Table 11 compares the delivered solution against the outcomes agreed at the scope stage. Six of the seven were met. The one that was not is publication of the dashboard to the Power BI Service, recorded here as unmet rather than described as complete.

Table 11. The delivered solution compared with the target outcomes.

| Requirement                                                                       | Target                                                      | Achieved                                       | Met |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------|-----|
| Model ranks cancellation risk better than chance and above the industry threshold | ROC-AUC at or above 0.80                                    | 0.8798                                         | Yes |
| Model identifies most bookings that actually cancel                               | Recall at or above 0.70                                     | 0.8226                                         | Yes |
| Flagged bookings are usually correct, to limit unnecessary guest contact          | Precision at or above 0.65                                  | 0.6857                                         | Yes |
| Model interpretable enough to explain the main risk drivers                       | Per-feature driver reporting available to a business reader | Yes, permutation importance on the test period | Yes |
| Full refresh completes within a working session on a laptop                       | End to end under 30 minutes                                 | 5 seconds to train the final model             | Yes |
| Drill down by hotel, market segment, channel, country and season                  | Available in the dashboard                                  | Built and verified against the control totals  | Yes |
| Dashboard reachable by URL through the Power BI Service                           | Published report                                            | Built in Power BI Desktop, not published       | No  |

The headline result is a ROC-AUC of 0.8798 on the held out period against a target of 0.80. Read plainly, shown one booking that cancelled and one that did not, the model gives the cancelled one the higher score about 88.0 percent of the time. Figure 4 shows the curves for all three optimized models on the same period.

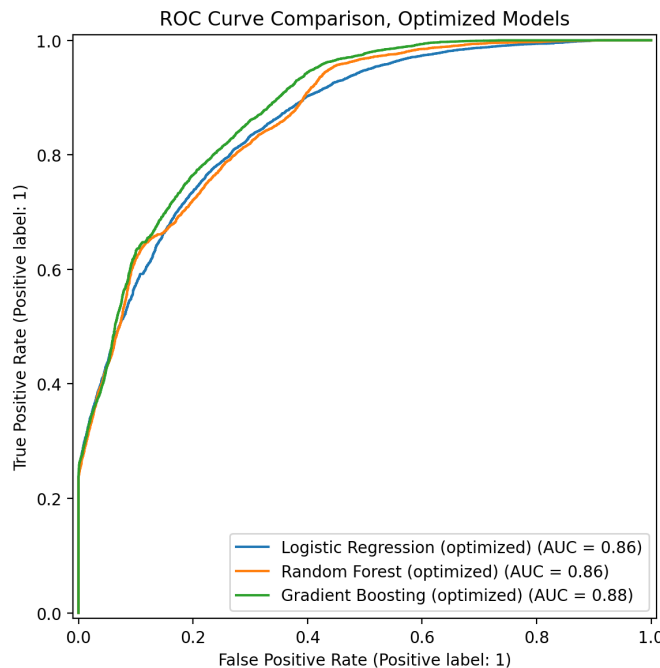


Figure 4. ROC curves for the three optimized models on the test period.

Figure 5 shows the confusion matrix at the selected threshold of 0.25. In business terms the solution flags 11,656 of the 23,713 bookings, which is 49.2 percent of the book. Of those, 68.6

percent really do cancel, so roughly three in ten flagged guests would be contacted unnecessarily, and of the bookings that really cancelled it catches 82.3 percent. That trade is deliberate and Section 10 explains it: a missed cancellation costs a room, an unnecessary contact costs a phone call.

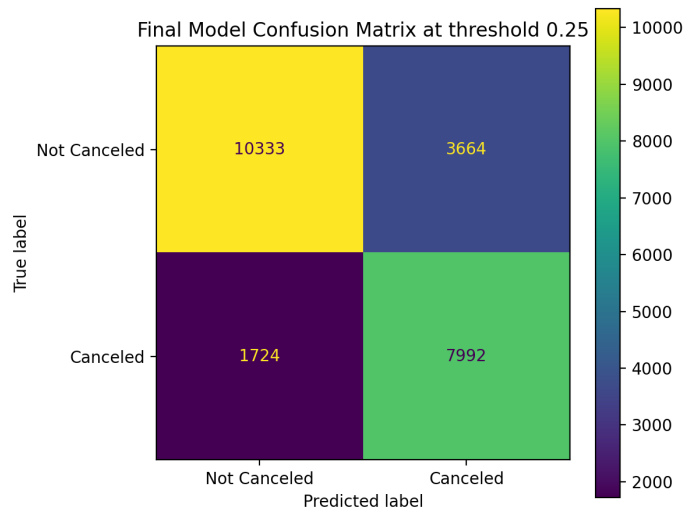


Figure 5. Confusion matrix for the final model at the operating threshold of 0.25.

### 9.3 Comparing the three approaches

Table 12 shows the three algorithms after optimization, all trained on the same reduced feature set and evaluated on the same period. The ranking is not the one the first implementation produced, which is the subject of Section 10.

Table 12. The three models after optimization, at the default 0.50 threshold.

| Model                           | ROC-AUC | Precision | Recall | F1     | Training time |
|---------------------------------|---------|-----------|--------|--------|---------------|
| Gradient Boosting (optimized)   | 0.8798  | 0.8191    | 0.6173 | 0.7040 | 5 s           |
| Random Forest (optimized)       | 0.8644  | 0.7740    | 0.6582 | 0.7114 | 6 s           |
| Logistic Regression (optimized) | 0.8598  | 0.7095    | 0.7532 | 0.7307 | 4 s           |

Gradient boosting was selected on ROC-AUC, the metric that matters for a solution whose job is to rank bookings. Logistic regression is close behind and remains useful as an explanation tool because its coefficients can be read directly. All three sit within 0.02 ROC-AUC of one another, which is worth stating plainly: the choice of algorithm mattered far less than the choice of features, which Section 10 quantifies.

## 9.4 Are the probabilities trustworthy

A ranking metric alone does not tell a revenue manager whether a probability of 0.70 means seven bookings in ten. Table 13 compares the mean prediction in each probability band with the proportion that actually cancelled.

*Table 13. Calibration of the predicted probabilities on the test period.*

| Probability band | Bookings | Mean predicted probability | Actual cancellation rate |
|------------------|----------|----------------------------|--------------------------|
| 0.0-0.2          | 9,616    | 0.068                      | 0.082                    |
| 0.2-0.4          | 6,006    | 0.267                      | 0.425                    |
| 0.4-0.6          | 1,532    | 0.499                      | 0.598                    |
| 0.6-0.8          | 4,152    | 0.708                      | 0.736                    |
| 0.8-1.0          | 2,407    | 0.986                      | 0.998                    |

The model is well calibrated at the two ends and understates risk in the middle. In the lowest band it predicts 0.068 against an actual 0.082, and in the highest 0.986 against an actual 0.998. In the 0.2 to 0.4 band it predicts 0.267 against an actual 0.425, so bookings there cancel more often than the number on the screen suggests. That is a limitation the revenue desk should know about, and it is one reason the operating threshold was set low enough to pull that band into the action list.

## 9.5 Business test scenarios

Metrics describe the whole test period. Table 14 traces three individual bookings of the kind a revenue manager would recognise through the full chain, from probability to recommended action.

*Table 14. Business test scenarios traced from probability to recommended action.*

| Scenario                                            | Probability | Risk band | Booking value | Revenue at Risk | Recommended action |
|-----------------------------------------------------|-------------|-----------|---------------|-----------------|--------------------|
| Scenario 1, low risk direct short lead booking      | 0.002       | Low       | \$190         | \$0.43          | Monitor            |
| Scenario 2, medium risk online travel agent booking | 0.362       | Watch     | \$560         | \$202.74        | Watch List         |
| Scenario 3, high risk long lead summer booking      | 0.991       | Very High | \$1,470       | \$1,456.76      | Request Deposit    |

The three scenarios behave as the design intends. A short lead direct booking scores near zero and is left alone, an online travel agent booking with a moderate lead time lands in the watch

band, and a long lead summer booking with no deposit scores close to certainty, triggers a deposit request, and because its booking value is high it sorts to the top of the queue.

## 9.6 Linking the output to the business requirement

Table 15 connects the model to the money. It shows how the 23,713 scored bookings distribute across the four risk bands, the exposure each band carries, and the proportion of each that actually cancelled.

*Table 15. Risk bands, recommended actions and Revenue at Risk on the test period.*

| Risk band | Recommended action | Bookings | Revenue at Risk | Actual cancellation rate |
|-----------|--------------------|----------|-----------------|--------------------------|
| High      | Retention Offer    | 4,017    | \$1,629,574     | 0.724                    |
| Very High | Request Deposit    | 3,306    | \$1,316,173     | 0.934                    |
| Watch     | Watch List         | 4,333    | \$700,561       | 0.460                    |
| Low       | Monitor            | 12,057   | \$629,650       | 0.143                    |

The bands separate cleanly, which is what makes the recommended actions defensible. The very high band cancels at 0.934 and the low band at 0.143, more than six times apart, so a revenue manager who works only the very high band is working a population where nine bookings in ten will cancel. Across all bands the solution measures \$4,275,959 of Revenue at Risk against a total booking value of \$12,101,899, and the action list of 11,656 bookings carries \$3,646,308 of it.

Figure 6 shows which fields the model relies on, measured by permutation importance on the test period rather than read off the training run. Country and deposit type lead, followed by agent, market segment and special requests. Lead time matters but ranks below the commitment signals, which is a useful finding for the business: how a booking was made says more about whether it will survive than how far ahead it was made.

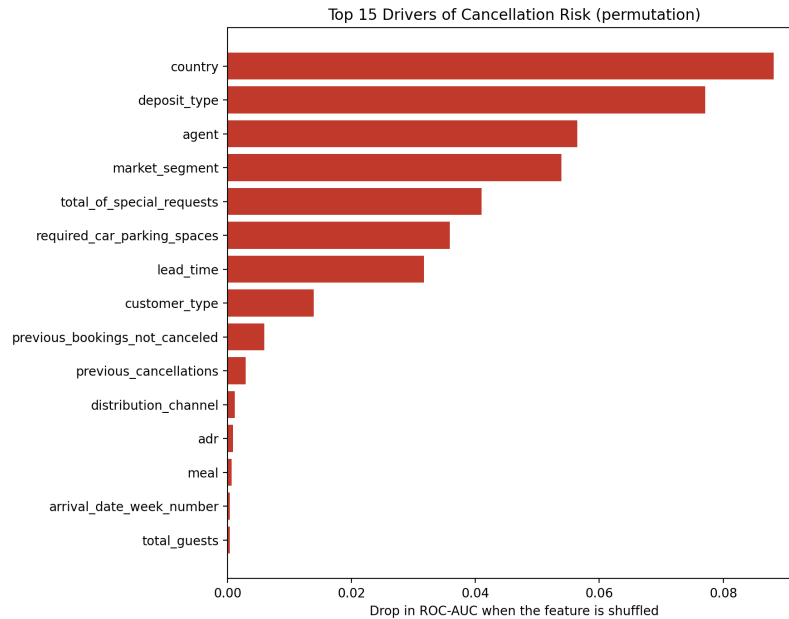


Figure 6. The strongest drivers of cancellation risk in the final model, by permutation importance.

## 9.7 Testing the reporting layer

The reporting layer was tested separately from the model, and this is where the most useful failure of the project was found. Nine control totals were computed from the scoring output and recomputed from the dashboard's own cached data. The first version failed three of them. Table 16 lists the values the reporting layer has to reproduce.

Table 16. Dashboard control totals computed from the scoring output.

| Measure                          | Definition                                             | Expected value |
|----------------------------------|--------------------------------------------------------|----------------|
| Total Bookings                   | COUNTROWS(Bookings)                                    | 23,713         |
| Predicted Cancellations          | Bookings[predicted_cancel] = 1                         | 11,656         |
| Predicted Cancellation Rate      | Predicted Cancellations / Total Bookings               | 49.2%          |
| Actual Cancellations             | Bookings[is_canceled] = 1                              | 9,716          |
| Actual Cancellation Rate         | Actual Cancellations / Total Bookings                  | 41.0%          |
| Total Booking Value              | SUM(Bookings[booking_value])                           | \$12,101,899   |
| Total Revenue at Risk            | SUM(Bookings[Revenue at risk])                         | \$4,275,959    |
| Revenue at Risk (Action List)    | Total Revenue at Risk filtered to predicted_cancel = 1 | \$3,646,308    |
| Average Cancellation Probability | AVERAGE(Bookings[cancel_probability])                  | 35.1%          |

The first build loaded 9,739 bookings where the delivered model flags 11,656, so it was reading a copy of the action list that predated the final optimization run. Its headline count ignored every slicer, because filter clearing had been added to stop two tables from disagreeing. And its

cancellation rate divided a test period numerator by a full history denominator, reading as roughly 8 percent against an actual 40.97%.

All three faults trace to one decision: the report had been assembled from two tables covering different populations. It was rebuilt on the single scored booking table, and every one of the nine measures and all four risk band totals now reconcile against Table 16. This is recorded here rather than quietly corrected, because a test that finds a fault is stronger evidence than a test that finds nothing, and because the fault was invisible on screen: the dashboard looked entirely reasonable while reporting a cancellation rate that was wrong by a factor of five.

Figure 7 and Figure 8 show the rebuilt dashboard. With no filter applied the overview page reports 23,713 bookings, 11,656 predicted cancellations and \$4,275,959 of Revenue at Risk, which matches the control totals exactly.

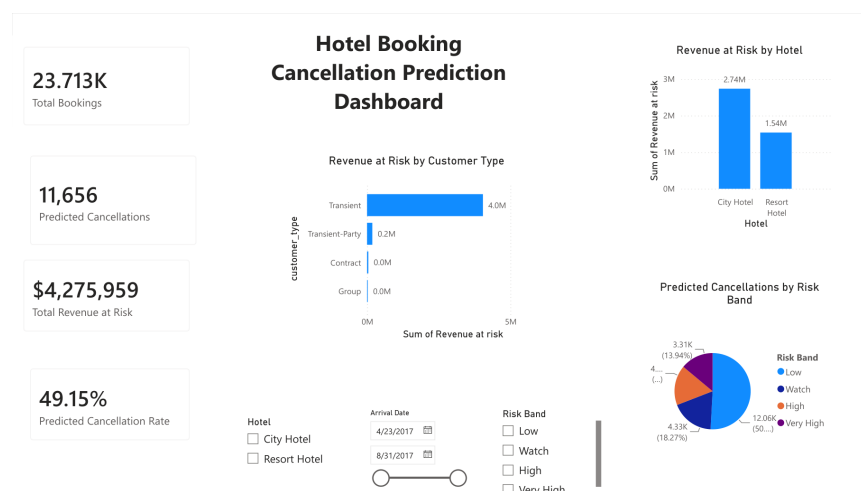


Figure 7. The dashboard overview page: indicator cards, Revenue at Risk by hotel and by customer type, and the risk band distribution.

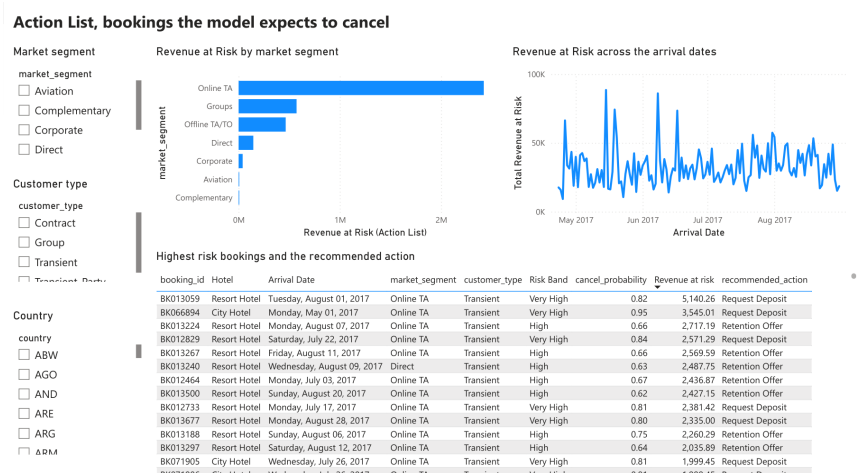


Figure 8. The action list page: Revenue at Risk by market segment, the trend across arrival dates, and the ranked table of flagged bookings with the recommended action.

The overview page answers the descriptive and diagnostic questions from Section 4 through its slicers, and the action list page answers the prescriptive ones. A revenue manager opens the second page, sorts by Revenue at Risk, and works down the list. That is the daily workflow the project exists to produce.

## 10. Potential Solution Optimization

The first working version of the solution was not the one delivered. Five optimization passes separate them, described below in the order they were carried out. The first produced by far the largest improvement, which is itself one of the findings of the project.

### 10.1 Removing a feature that could not generalise

The first implementation included the arrival year and month as a model feature. On cross validation it looked helpful. On the held out period it was actively harmful, and the reason is visible in the split: training covers 22 months and the test period covers 5, of which 4 were never seen during training (2017-05, 2017-06, 2017-07, 2017-08). The model had learned month specific behaviour that did not exist in the future it was asked to predict.

The size of the gap between cross validation and the test period was the signal. Gradient boosting scored 0.8771 on cross validation and 0.8238 on the test period, a gap of 0.0533. Removing the calendar feature closed most of it. Table 17 shows what that single change did to each algorithm.

Table 17. Effect of removing the calendar feature on each algorithm.

| Algorithm           | ROC-AUC before | ROC-AUC after | Change  | Time before | Time after |
|---------------------|----------------|---------------|---------|-------------|------------|
| Logistic Regression | 0.8570         | 0.8598        | +0.0028 | 114 s       | 4 s        |
| Random Forest       | 0.8562         | 0.8644        | +0.0082 | 176 s       | 6 s        |
| Gradient Boosting   | 0.8238         | 0.8798        | +0.0560 | 177 s       | 5 s        |

Gradient boosting gained 0.0560 ROC-AUC and moved from the weakest of the three models to the strongest, while ten rounds of hyperparameter search across both tree ensembles achieved almost nothing next to it. The lesson the team would carry into any future project is that time spent asking whether a feature can exist in the future is worth more than time spent tuning.

### 10.2 Fixing a convergence failure

The logistic regression baseline had been reporting a convergence warning, which was noted and not acted on. Investigating it showed the solver was stopping before it had finished fitting. Changing the solver and using the reduced feature set fixed it, and the model now converges in 191 iterations, moving ROC-AUC from 0.8570 to 0.8598. The accuracy gain is small, but training time fell from 114 seconds to 4 seconds, which changes how often the team could afford to rerun the pipeline, and a model that has not converged is not one whose coefficients can be quoted as risk drivers.

### 10.3 Tuning the decision threshold

A classifier defaults to calling anything above 0.50 a cancellation, but 0.50 has no business meaning, and the two errors are not symmetric: a missed cancellation costs an empty room that could have been resold, while an unnecessary flag costs a phone call. The threshold was therefore treated as a business parameter rather than a default. Table 18 shows precision and recall across the range, and Figure 9 shows a modelled net business value using the assumptions stated in Section 5.

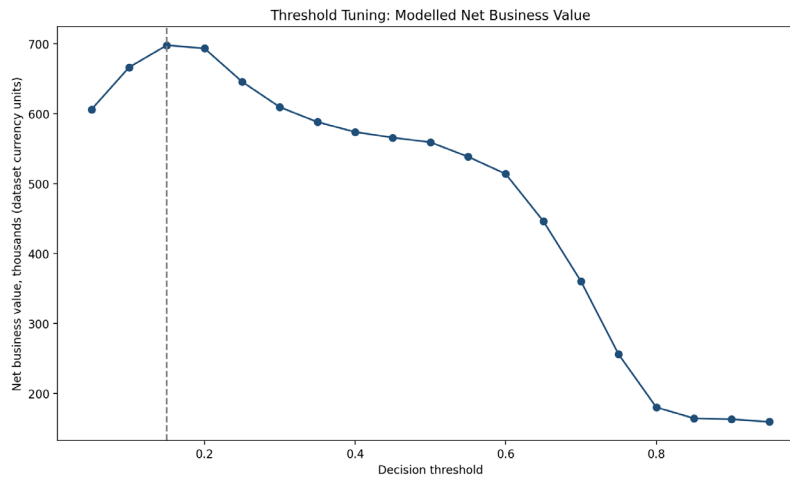


Figure 9. Modelled net business value across the threshold range.

Table 18. Selected points from the threshold sweep.

| Threshold | Bookings flagged | Precision | Recall | F1    | Net modelled value |
|-----------|------------------|-----------|--------|-------|--------------------|
| 0.20      | 14,097           | 0.633     | 0.918  | 0.750 | \$693,213          |
| 0.25      | 11,656           | 0.686     | 0.823  | 0.748 | \$645,509          |
| 0.30      | 9,323            | 0.751     | 0.720  | 0.735 | \$609,233          |
| 0.50      | 7,323            | 0.819     | 0.617  | 0.704 | \$558,993          |
| 0.70      | 5,035            | 0.852     | 0.441  | 0.581 | \$360,094          |

The threshold was set to 0.25. Modelled net value rises from \$558,993 at the default 0.50 to \$645,509, an improvement of \$86,516 on the test period alone and with no retraining. The unconstrained best F1 sits lower still at 0.2, but that point drops precision to 0.633, which would send more than a third of the contacts to guests who were never going to cancel. The team chose the point that satisfies the agreed precision floor rather than the one that maximises a single metric.

#### 10.4 Realigning the business rules with the new threshold

Moving the threshold broke the risk bands, which had been drawn at the design stage assuming a 0.50 threshold. That left a gap where the model predicted a cancellation but the business rules still said monitor. Table 19 states the rebased values, so that the first action band begins where the model starts predicting a cancellation. The four band structure and the four actions are unchanged from the design; only the cut points moved.

Table 19. Risk bands rebased on the selected operating threshold.

| Risk band | Probability range | Recommended action |
|-----------|-------------------|--------------------|
| Low       | Below 0.25        | Monitor            |
| Watch     | 0.25 to 0.50      | Watch List         |
| High      | 0.50 to 0.75      | Retention Offer    |
| Very High | 0.75 and above    | Request Deposit    |

## 10.5 Rebuilding the reporting layer

The fifth pass was the dashboard rebuild described in Section 9.7. It counts as optimization rather than a defect fix because the change was structural: the report was rebuilt on one table instead of two. Once the two populations were collapsed into one, the filter clearing could be removed and the indicator cards began responding to the slicers as a user would expect.

## 10.6 What was tried and rejected

Hyperparameter search across both tree ensembles ran for ten rounds and produced almost no improvement compared with the feature change. It was kept because the selected parameters are defensible and the search cost little once training time had dropped, but it is not presented as the source of the result. Oversampling the minority class was rejected because roughly four bookings in ten cancel and the classes are not badly imbalanced, and adding the raw company identifier back was rejected because it is 94 percent empty.

## 10.7 What remains to be optimized

Four opportunities remain, and they are listed in the order the team would take them.

- Publish the dashboard to the Power BI Service and schedule the refresh. This is the one target outcome that was not met, and it is the difference between a report a team member opens and one the revenue desk uses.
- Replace the file handoff with a scheduled export from the property management system, so the solution scores the forward book each night rather than a fixed historical period.
- Calibrate the middle probability band, so the dashboard can quote a probability a revenue manager reads literally.

- Close the feedback loop. Recording which actions were taken and whether the booking survived would replace the assumed save rate in Section 5 with a measured one, and would supply labelled outcomes for retraining.

## 11. Conclusion

The project set out to answer a question a hotel cannot currently answer: which of the bookings on the books are likely to cancel, while there is still time to do something about it. It delivered a working answer. A gradient boosting model reaches a ROC-AUC of 0.8798 on a held out period it never saw during training, catches 82.3 percent of the bookings that really cancel, and is right about 68.6 percent of the bookings it flags. Those predictions become \$4,275,959 of measured Revenue at Risk and a ranked action list of 11,656 bookings, each carrying a risk band and a recommended action.

Two findings are worth carrying beyond this report. The largest single improvement came from removing a feature that could not generalise, not from tuning the model. And the most serious fault in the delivered solution was in the reporting layer rather than in the model, was found by a control total test, and was invisible to anyone looking at the screen. Both point the same way: the value of a business intelligence solution depends on the whole chain being correct, and the parts that look least interesting are the ones worth testing hardest. The recommendation to the business is to adopt the ranked action list as the daily working queue for the revenue desk, starting with the very high band, where more than nine bookings in ten cancel. Before the recommended actions are trusted operationally the model should be retrained on current data from the hotel's own property management system, and the dashboard should be published so that the people who make the decisions can reach it without installing anything. The value delivered is that a revenue manager can now open one page, see where the money is exposed, and know which reservations to work first.

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## Appendix A. Business Glossary

The terms below are used throughout the report and in the dashboard, and they are defined here so that a reader from the business side and a reader from the technical side use them the same way.

*Table 20. Business glossary.*

| Term                     | Definition                                                                   |
|--------------------------|------------------------------------------------------------------------------|
| ADR                      | Average Daily Rate, the average room rate associated with a booking          |
| Lead time                | The number of days between the booking being made and the scheduled arrival  |
| OTA                      | Online Travel Agent, a third party digital booking intermediary              |
| PMS                      | Property Management System, the operational hotel reservation system         |
| Booking value            | ADR multiplied by the number of booked room nights                           |
| Cancellation probability | The model estimated likelihood that a booking will be cancelled              |
| Revenue at Risk          | Booking value multiplied by cancellation probability                         |
| Operating threshold      | The probability above which a booking is flagged as a predicted cancellation |
| ROC-AUC                  | The chance the model scores a cancelled booking above one that was honoured  |
| Precision                | The share of flagged bookings that really cancelled                          |
| Recall                   | The share of real cancellations that the model flagged                       |
| Schema contract          | The fixed agreement on field names, order and types that the dashboard reads |

## Appendix B. Individual Contribution Log

Contributions are recorded against the project work breakdown structure. Each entry reflects work that was completed and verified rather than work that was assigned.

*Table 21. Individual contribution log across the four project stages.*

| Team member     | Role                                   | Contribution across the project                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Asaad Razeq     | Project Manager and Documentation Lead | Coordinated the four stages and the submission schedule. Built the exploration, cleaning and export scripts. Converted the modelling notebook into a reproducible pipeline. Diagnosed the convergence failure, identified and removed the non generalising calendar feature, added the threshold tuning and the business value model, built the control total test that found the reporting fault, rebuilt the Power BI handoff, and compiled every written deliverable including this report. |
| Said Huner      | Data and Machine Learning Lead         | Built the solution development notebook: the temporal split, the dual preprocessing pipelines, the evaluation function, the three initial models, the randomised search tuning of both tree ensembles, the confusion, ROC and precision recall diagnostics, the prediction and action list outputs, and the three business test scenarios. Produced the design stage diagrams and reviewed the technical accuracy of the written sections.                                                     |
| Nora Yong       | Data and Machine Learning Support      | Paired with the Data and Machine Learning Lead. Verified the model ready feature list against the Assignment 2 data dictionary, which is how the leakage and field availability rules were confirmed. Reviewed the evaluation results and the test scenario outputs, checked formatting and APA references, and maintained the run results and contribution records.                                                                                                                           |
| Cassidy Taillon | Visualization Lead                     | Designed and implemented the Power BI dashboard: imported and prepared the scoring output, created the DAX measures and the indicator cards, built the charts and the risk band distribution, added the interactive slicers, and organised the two report pages. Tested the visuals and the filter behaviour, and authored the stakeholder analysis and the dashboard facing content of the written deliverables.                                                                              |